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Integrated Physical Pharmacy I

Laboratory Manual

Physical pharmacy I (Phar 212)

Laboratory Manual

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EXPERIMENT I

SOLUBILITY OF PARTIALLY MISCIBLE LIQUIDS

Objective:

To investigate the effect of concentration and temperature on miscibility of phenol-water system and to determine the critical solution temperature.

Background:

In most cases, partially miscible liquids become more soluble in each other with increasing temperature until the critical solution temperature is reached, above which there is complete miscibility. In this experiment mixtures of known composition of Phenol and water are heated until they dissolve completely. This is easily recognized by the disappearance of cloudiness due to two phases in increase temperature and reappearance on cooling. When the percent composition is plotted against temperature of miscibility a curve is obtained with a maximum at a critical solution temperature.

Procedure:

A set of test tubes is made up containing the following quantities of phenol and water.

Test tube No.	1	2	3	4	5	6	7	8
Phenol (Gm)	0.4	0.8	1.0	1.2	1.4	1.6	2.0	2.4
Water (Cm ³)	3.6	3.2	3.0	2.8	2.6	2.4	2.0	1.6

The test tubes are plugged and supported, one at a time, in a holder made of wire suspended in a beaker of water. The beaker is placed on water bath, and contains a thermometer reading to 1°C. By warming the water in the beaker on water bath and by shaking the test tube while it is suspended in the water, the temperature at which the cloudy liquid just goes clear is found. The temperature is also found for reappearance of the cloudiness on cooling and the mean of the two temperatures is taken as the temperature of miscibility for that mixture. The procedure is repeated for each test tube, and a graph is plotted taking temperature of miscibility against percentage composition for the various mixtures. This gives a curve, the maximum being the critical solution temperature. (Literature value = 66.8°C).

Results and Discussion:

Test tube Number	Phenol (g)	Water (cm ³)	Percentage Composition	Clearing Temp. (°C)	Cloudy Temp. (°C)	Average Temp. (°C)
1	0.4	3.6	10			
2	0.8	3.2	20			
3	1.0	3.0	25			
4	1.2	2.8	30			
5	1.4	2.6	35			
6	1.6	2.4	40			
7	2.0	2.0	50			
8	2.4	1.6	60			

1. Draw the composition-temperature graph and determine the critical solution temperature of the phenol water system. Compare the experimentally determined result with the literature value.
2. Discuss the pharmaceutical application of this experiment.

EXPERIMENT II

THREE COMPONENT SYSTEM

Objective:

To study a three component system: Carbon tetrachloride, Acetic acid and Water.

Background:

Pharmaceutical preparations, for a multitude of reasons, usually consist of two or more components. Many of these preparations are combinations of immiscible materials to which a blending agent must be added to produce a clear homogenous solution.

'Blending' is a term used to describe the mutual solubility of normally immiscible liquids in the presence of an amphiphilic agent. Controlling the amount of blending agent used is an important task. An insufficient quantity provides "Layering" which can be extremely dangerous in pharmaceutical preparations and excessive blending agent is undesirable due to side effects.

The three component systems is best described and interpreted by means of a phase rule and phase diagram. At constant temperature and pressure, Gibb's phase rule for a three component system is given by the following equation:

$$F = C - P$$

Where F is degree of Freedom, P is the number of phases at equilibrium, and C is number of components comprising the system. The degree of freedom is defined as the least number of variable factors (e.g., temperature, pressure and concentration, refractive index, viscosity etc.) which must be arbitrarily fixed to have the condition of the system perfectly defined.

In order to assure a symmetrical representation of the three liquids in a ternary system an equilateral diagram is employed. The following rules should be taken into account in equilateral triangle diagram method.

1. In an equilateral triangle the sum of the distances from any given point to the three sides, along the perpendiculars to the sides, is equal to the height of the triangle.
2. Each apex of the equilateral triangle represents 100 percent by weight of one component and zero for the others.
3. Each side is divided into 100 equal parts and the distance from each apex to the center of the opposite side of the equilateral triangle is divided into 100 equal parts.
4. The area within the triangle represents all the possible combinations of the three components.

A very important feature of the phase diagram is the tie line, which connects the compositions of the two phases, which are in equilibrium. The relative amounts of the two phases can be calculated by measuring the distance along the tie line from point representing the gross composition of the mixture to the points representing the two solutions at equilibrium. Usually, the tie lines are not parallel to each other or to a side of the triangle. The only case in which the tie lines are parallel is when the blending agent has the same degree of solubility in either of the immiscible substances. In carbon tetrachloride, water and acetic acid system, carbon tetrachloride and water are immiscible liquids and acetic acid is the blending agent.

Procedure:

The most accurate way to determine the three component system of carbon tetrachloride, glacial acetic acid and water is first to prepare nine 25 g samples of the two organic liquids which are miscible. The following percentages of w/w are suggested 10,20,30,40,50,60,70,80, and 90. A volumetric procedure could be employed with the aid of density values. Titrate each sample while shaking with distilled water until the appearance of turbidity. Although turbidity does not necessarily indicate a distinct separation into two phases, it suggests an unstable state that will probably layer with time into two distinct phases.

Upon completion of titration, signaled by turbidity of the system, the samples under investigation no longer weight 25g. The exact weight for practical consideration is 25g plus the volume of water added to produce turbidity (since 1 ml. weighs approximately 1g). The quantity of each component in the samples will be calculated (% w/w) and the system will be plotted.

Discussion:

1. Explain the pharmaceutical importance of determining the three component system.
2. Discuss tie lines of equilateral diagram and their application.

EXPERIMENT III

DETERMINATION OF HLB VALUE OF A GIVEN SURFACTANT

Objective:

To determine the hydrophilic-lipophilic balance (HLB) value of a polyhydric alcohol fatty acid esters (surfactants).

Background:

Surfactants (Surface active agents) are molecules or ions that are adsorbed at interfaces or surface and cause a marked decrease in the interfacial tension or surface tension. A surfactant concentrated or accumulated at interfaces because those molecules or ions have a certain affinity for both polar and non-polar solvents. For a molecule or ion to have an affinity to both polar and non-polar solvent, its chemical structure has to have a portion of both the hydrophilic (water liking) and lipophilic (lipid-liking) moiety. Surfactants are characterized by the possession of such distinct regions in their chemical structure. The existence of such moieties in a molecule is referred to as amphipathy and therefore, surfactants are also called amphiphiles. Depending on the number and nature of hydrophilic and lipophilic groups present, the surfactant may be predominantly hydrophilic, lipophilic or be reasonably well balanced between these two extremes.

Griffin (1949) devised an arbitrary scale of values in which each surfactant could be assigned a number as a measure of its hydrophilic-lipophilic balance (HLB). Thus by means of this system, it was possible to classify surfactants into different classes of application.

Function of surfactants and HLB values

<u>HLB values</u>	<u>Agents</u>
1-3	Anti-foaming agents
3-8	W/o emulsifying agents
8-16	O/w emulsifying agents
16-19	Solubilizing agents
7-9	Wetting and spreading agents
13-16	Detergents

Along the scale lipophilic properties of the surfactant decreases as the scale number increases and the hydrophilicity increases with increasing the scale value. The spans, sorbitan esters, are lipophilic and have low HLB values (1.8 to 8.6). The tweens, polyoxyethylene derivatives of the spans, are hydrophilic and have high HLB values (9.6 to 16.7).

The HLB of a number of polyhydric alcohol fatty acid esters, such as glyceryl monostearate could be estimated by using the following equation.

$$HLB = 20(1-S/A) \text{ ----- eq. 1}$$

Where S is the saponification number of the ester and A is the acid number of the fatty acid.

Procedure

Determination of saponification value

Weigh accurately about 2gm of the surfactant (glyceryl monostearate) in a flask and add 25ml of an alcoholic solution of potassium hydroxide (0.5N KOH in 95% alcohol). Attach a reflux condenser, and boil for one hour frequently rotating the contents. While the solution is still hot, add 2 drops of phenolphthalein solution and titrate the excess of alkali with 0.5N hydrochloric acid. Carry out the blank titration. The saponification value is calculated using the following equation.

$$\text{Saponification value} = \frac{(b-a) \times 0.02805 \times 1000}{W} \text{ ----- eq. 2}$$

Where 'a' is the volume of HCl required and 'b' is the volume of HCl consumed during the blank titration. 'W' is the weight of the surfactant used. The HLB value of the surfactant is then calculated using equation 1.

Acid Value

The acid value of the fatty acid and the theoretical HLB value of some surfactants:

Surfactant	Acid value of the fatty acid	HLB value (Literature value)
Span 20	278.9	8.6
Span 40	218.9	6.7
Span 60	207.5	4.7
Span 80	198.6	4.3
Tween 20	278.9	16.7
Tween 40	218.9	15.6
Tween 60	207.5	14.9
Tween 80	198.6	15.0
Glyceryl mono stearate	207.5	3.8

Discussion

Explain the following

1. Why the blank titration is required in this experiment
2. Define saponification number and acid value
3. Discuss the pharmaceutical application of interfacial tension.

EXPERIMENT IV

ADSORPTION ISOTHERM

Objective:

To determine the adsorption of acetic acid from a solution on activated charcoal at constant temperature.

Background:

Adsorption is a surface phenomenon. It consists of an interaction between the moving molecules, which could be gas, liquid or solution, called adsorbate and a relatively fixed surface, usually solid, called adsorbent. The degree of adsorption of a liquid by a solid depends on a number of factors, such as: chemical nature of the adsorbent and the adsorbate, the surface area of the adsorbent, temperature and concentration of the adsorbate. Depending on the chemical nature of the adsorbate and the adsorbent, the adsorption could be of chemical adsorption, where the adsorbate is attached to the adsorbent by primary chemical bond (chemisorption) or physical adsorption, where the adsorbate attached to the adsorbent by Van der Waals force (weak force), therefore, adsorption is reversible.

The chemical adsorption is irreversible, i.e. desorption is not possible. Whereas, desorption could be achieved in physical adsorption by increasing temperature and/or decreasing concentration. This could mean, in physical adsorption the temperature and equilibrium concentration of the adsorbate affect the degree of adsorption.

The adsorbed molecules may come from a gas phase, a liquid, or from a solution in a liquid. Adsorption from solution is usually monomolecular. In this case, adsorption ceases when the surface is completely covered with a layer one molecule thick. Adsorbed layers more than one molecule thick has been proved to exist in certain cases, yet they are rare.

The amount of adsorbate adsorbed on the adsorbent is a function of both temperature and concentration. If we keep the temperature constant (isotherm), the degree of adsorption varies with the concentration of the solution resulting in adsorption isotherm.

This relationship between the amount adsorbed and the concentration of the solution, for monomolecular adsorption, is empirically explained by the following two equations.

1. Freundlich equation

$$X/M = KC^{1/n} \quad \text{or} \quad \log (X/M) = \log K + 1/n \log C \dots\dots\dots \text{eq. 1}$$

Where X/M is the weight of the adsorbed material (X) per gram of adsorbent (M), C is the concentration of the solution at equilibrium and K and n are constants to be evaluated from experimental data.

Freundlich equation is derived on assumption that adsorption ceases when the surface of adsorbent is covered with layer of one molecule thick (monomolecular adsorption).

2. Langmuir equation

$$\frac{X}{M} = \frac{abc}{1 + ac} \quad \text{or} \quad \frac{C}{X/M} = \frac{1}{ab} + \frac{C}{b} \dots\dots\dots \text{eq. 2}$$

Where X/M and C are the same as above, "a" is a constant proportional to the heat of adsorption and the temperature and "b" is the amount of adsorption when the surface is covered with a monomolecular film.

Langmuir equation has a sound theoretical background. It considers a proportional constant of the heat of adsorption, the temperature and the amount of adsorption when the surface is covered with a monomolecular film. However, Langmuir equation is based on the assumption that the solid surface is completely uniform. Deviations from the equation indicate lack of uniformity in the surface.

Procedure:

Prepare carefully, 120 ml of N/4, N/8, N/16, N/32 and N/64 solutions of acetic acid, by successive dilution, starting from the stock solution (N/2 acetic acid solution). Weigh accurately 2 g of activated charcoal (previously pulverized and dried) each into six labeled flasks and pipette 100 ml of the above acetic acid solution. Note the concentration of the acetic acid solution and label the flask. Plug the flasks and set-aside in thermostat with frequent stirring, for about 1 hr to allow equilibrium. If the room temperature is reasonably constant ($\pm 2^\circ\text{C}$.), a thermostat may not be necessary.

Meanwhile, titrate three 10 ml aliquots of the original N/2 acetic acid with 0.1N NaOH, using phenolphthalein as an indicator. If the dilutions were done carefully, their concentrations may be calculated from the standardized N/2 solution. Otherwise, the original concentration of each dilution should be determined by titration.

After the equilibrium is reached, take at least two samples from each flask with a filtering pipette and titrate the acid with 0.1N NaOH, using phenolphthalein as an indicator. 10ml samples are suggested for the more concentrated solutions and 25 ml for the dilute ones. However, if the volume of base required for the first sample is too small, a larger aliquot should be taken for the second.

Results:

Standardization of acetic acid solution

	Volume of CH_3COOH	Volume of NaOH	Conc. Of NaOH	Conc. Of CH_3COOH
Test No 1				
Test No 2				
Test No 3				
AVERAGE				

Adsorption of acetic acid on charcoal

Conc. of Acetic acid (N)	Weight of charcoal (gm)	Original conc. of Acetic acid (C ₁)	Final conc. Of Acetic acid (C ₂)	Amount of acetic acid adsorbed in grams (x)
(X)				
N/2				
N/4				
N/8				
N/16				
N/32				
N/64				

1. Plot x/m Vs C_2 and estimated the value of x/m for monomolecular coverage (i.e. saturation value).
2. Plot $\log x/m$ Vs $\log C_2$, draw a smooth curve through the points and estimate the Freundlich's constants k and n .
3. Plot $C_2/(x/m)$ Vs C_2 , draw the best straight line and calculate the constants 'a' and 'b' of Langmuir's equation.

Discussion

1. Why do we standardize the stock solution of acetic acid?
2. Discuss the pharmaceutical application of adsorption?

EXPERIMENT V

INFLUENCE OF TEMPERATURE ON SOLUBILITY OF DRUGS

Objective:

To study the effect of temperature on solubility of slightly soluble drugs, using benzoic acid in distilled water, and calculate the differential heat of solution (ΔH) of the system.

Background:

The solubility of a drug is defined as the concentration of the solute in solution which is in dynamic equilibrium with the solid phase at a given temperature. It is the amount of the drug in a saturated solution.



The influence of temperature on solubility depends on the differential heat of solution which is defined as the heat absorbed when one mole of solute is dissolved in such a quantity of the solution that no appreciable change in the concentration takes place. Generally heat is absorbed when a majority of pharmaceuticals dissolved (endothermic process) and they are more soluble at an elevated temperature. This effect is in accordance with the Le chatelier principle, which states that, if a system in equilibrium is disturbed, the system will shift in such a way as to counteract the effect of disturbance. Increasing the temperature of a system could be considered as a state of disturbance.

Conversely, if heat is evolved during the process of dissolution (exothermic process), the solubility will decrease with an elevation of temperature. Dissolution of methylcellulose and calcium hydroxide is an exothermic process.

A saturated solution in contact with solid drug is one of the simplest cases of equilibrium, in that molecules leave the solid and pass into solution at the same rate at which the molecules from the solution are deposited on the solid. If the drug forms an ideal solution, the influence of temperature on the solubility of such system can be described mathematically by the following two forms of Van't Hoff's equation.

$$\text{Log } S = \frac{-\Delta H}{2.303R} \cdot \frac{1}{T} + \text{constant}$$

$$\text{Log } \frac{S_2}{S_1} = \frac{-\Delta H}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Where S_2 and S_1 are the solubilities in moles/liter or in moles/Kg of solvent at absolute temperatures T_2 and T_1 . ΔH is the average differential heat of solution over the temperature

ranges (cal/mole). R is the gas constant expressed in calories per mole per degree ($R=1.987$ Cal/mole deg) or (0.08205 liter atm/mole deg).

If the differential heat of solution were constant and there were no complicating reaction, such as ionization of drug, a plot of $\log S$ Vs $1/T$ would yield a straight line with a slope of $-\Delta H/2.303R$, from which ΔH can be calculated.

The general procedure consists of equilibrating an excess amount of a drug with a given volume of solvent at a particular temperature. In this experiment, distilled water in a beaker is saturated with benzoic acid by dissolving the crystals at a higher temperature and then cooling to the desired temperature so that some of the dissolved drug crystallized out. When we approach the system from a side of super saturation the equilibrium is attained rapidly with good precision. At equilibrium, an aliquot portion of supernatant solution is withdrawn and assayed for the drug content.

Procedure:

One gram of Benzoic acid is placed in a wide mouthed Erlenmeyer flask containing about 100ml. Of distilled water. The flask is then heated on a hot plate until the drug is completely dissolved. The solution is removed from heat and allowed to cool slowly with constant and careful stirring with a stirring thermometer.

When the temperature drops to about 55° , the flask is placed in a larger beaker. The air jacket between the flask and the beaker provide a slower rate of cooling. When the temperature reaches about 50° , 10 ml of an aliquot portion of a supernatant solution is pipetted rapidly into 100 ml. Erlenmeyer flask containing 20 ml of distilled water. The exact temperature at the time sampling is recorded. A small piece of cotton containing tube is plugged to the tip of the pipette to prevent withdrawal of any solid. The filtering device (rubber tube clogged with cotton) is removed before draining the pipette. Similarly samples are withdrawn at 40° , 30° , 20° and 10° . It is necessary to warm the pipette by running hot water over the outside of the pipette before removing the sample. The temperatures of 20° and 10° are attained by using cold water and ice bath.

Two drops of phenolphthalein indicator solution is added to each of the above solutions and titrated with 0.05N NaOH. The appearance of faint pink colour is the end-point marker. The amount of sodium hydroxide consumed is recorded and the differential heat of solution is determined.

Results:**Solubility data of Benzoic acid**

Temperature (°C)	Vol. Of 0.05N NaOH (ml)	Solubility (G/100 ml)	Solubility (S) (Moles/liter)	Log S	1/T
50					
40					
30					
20					
10					

ΔH from plot =

ΔH from literature =

Plot log S Vs $1/T$ and draw the best straight line through the points and calculate the differential heat of solution of benzoic acid.

EXPERIMENT VI

SOLUBILITY PRODUCT

Objective:

To introduce the basic techniques of determination of the solubility product using potassium hydrogen tartrate as an example.

Background:

When slightly soluble electrolytes are dissolved to form saturated solutions, the solubility is described by a special constant, known as the solubility product K_{sp} of the compound.

Potassium hydrogen tartrate, which may be represented as KHTa is a sparingly soluble salt which breaks up in solution to give potassium ions and weakly acidic hydrogen tartrate ions.



$$K_{sp} = [\text{K}^+] [\text{HTa}^-]$$

The hydrogen tartrate ion is such a weak acid that its dissociation into hydrogen and tartrate ions may be neglected. The concentration of hydrogen tartrate ions in solution may be determined readily by titration against standard alkali using phenolphthalein as indicator. Addition of another salt containing potassium ion reduces the solubility of potassium hydrogen tartrate because of the common ion effect but it should be found that K_{sp} remains constant.

The principle of reducing the concentration of sparingly soluble substance by the common ion effect has many applications, particularly in quantitative and qualitative analysis. Thus in the gravimetric determination of sulphate ion by precipitation as the barium salt the precipitate is washed with the dilute sulphuric acid solution rather than with water.

If the pharmacist wishes to prevent precipitation of slightly soluble salt in water, he may add some substance which will tie up and reduce the concentration of one of the ions, more of the salt will then pass from the undissolved to the dissolved state until the solubility product constant is reached and the equilibrium reestablished. The solubility of iron compounds is increased by citrate due to complex formation of Fe^{+++} with sodium citrate.

Procedure:

About 1g of finely powdered KHTa is placed in each five test tubes. To the first distilled water is added the tube is nearly full. To the others the following four standard solutions of KCl are added: tube 2, 0.01 mol dm^{-3} tube 3; 0.02 mol dm^{-3} ; tube 4, 0.03 mol dm^{-3} tube 5, 0.04 mol dm^{-3} . The five test tubes are stoppered and well shaken for five minutes, and then allowed to occasional shaking for 20 min. At the end of this period, 5 cm^3 portions are withdrawn from the tubes by means of pipette fitted with a glass wool filter. (Glass wool is contained in a rubber connection between the end of the pipette and an extra jet drawn from glass tubing. The filter is removed before delivery the liquid from the pipette). Each portion is titrated with 0.02 mol dm^{-3} NaOH. From each titre, the concentration of Hta-1 in the corresponding solution is calculated. The concentration of K^+ is therefore the sum of this and the added concentration of KCl. The solubility product is obtained by multiplying the total K^+ concentration in each solution by the Hta^- concentration. It should be found that K_{sp} is approximately the same in all the solutions.

EXPERIMENT VII DETERMINATION OF DISSOCIATION CONSTANT

Objective:

To determine the dissociation constant of salicylic acid

Background:

Drugs are usually weak acids or weak bases. A weak acid or base is only slightly dissociated. Therefore, the fraction of the drug that is ionized in solution is given by the dissociation constant of drug. These constants are expressed in terms of pK_a for both acidic and basic drugs.

The equilibrium between unionized molecule of an acidic drug and the ions produced by the dissociation may be written as



The dissociation constant " K_a " for this drug is

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Where concentrations are expressed in moles per liter. Now taking logarithms and reversing the signs.

$$\begin{aligned} -\log K_a &= -\log [H^+] - \log [A^-] + \log [HA] \\ pK_a &= pH + \log \frac{[HA]}{[A^-]} \end{aligned}$$

A general equation for acidic drug with one ionisable group may be written as

$$pK_a = pH + \log \frac{C_u}{C_i}$$

Where C_u and C_i are concentrations of unionized and ionized molecule respectively.

Procedure:

Equimolar (0.1M) solutions of salicylic acid and sodium salicylate were prepared in 50 ml 30% 40% and 60% alcohol. Twenty ml from each sample of the same alcoholic concentrations was thoroughly mixed and pH of the mixture was determined. A graph was plotted taking pH (Y-axis) and the strength of alcohol used (X axis). The resulting straight line is extrapolated to Y-axis to obtain pK_a of salicylic acid.

Dissociation constant of some weak acids in aqueous solutions at 25°C is given below:

WEAK ACIDS	K_a	pK_a
Acetic acid	1.75×10^{-5}	4.76
Acetyl salicylic acid	3.27×10^{-4}	3.49
Barbituric acid	1.05×10^{-4}	3.98
Benzoic acid	6.30×10^{-5}	4.20
Boric acid	5.8×10^{-10}	9.24
Citric acid	8.7×10^{-4}	3.06
	$1.8 \times 10^{-5} (2H)$	4.74
	$4.0 \times 10^{-6} (3H)$	5.40
Salicylic acid	1.06×10^{-3}	2.997
Sulfadiazine	3.3×10^{-7}	6.48

EXPERIMENT VIII DISTRIBUTION

Objective:

The objectives of this experiment are to determine the distribution ratio of acetic acid between water and carbon tetrachloride and to study the degree of association or dissociation of acetic acid in the solvents.

Background:

When a solute is added to a mixture of two immiscible liquids and the system allowed to come to equilibrium at some given temperature, the following relationship will hold regardless of the quantity of solute added:

$$K_a = \frac{a_2^n}{a_1}$$

Where K_a is the distribution constant, a_1 and a_2 are the activities of the solute in the two solvents, and 'n' is the ratio of the average molecular weight in solvent 1 to its average molecular weight in solvent 2. Association of acetic acid will increase the average molecular weight, while dissociation will decrease it. It is to be noted that 'n' may vary considerably with concentration due to changes in the degrees of association and dissociation.

In order to simplify this experiment, concentration will be substituted for activities in the above equation, so that the distribution equation becomes:

$$K_c = \frac{C_2^n}{C_1}$$

If, for any distribution system, the values of C_1 and C_2 are obtained for two different total solute concentrations, both n and K can be calculated by solving the two equations for the two unknowns.

In carrying out these calculations it may be convenient to take the logarithm of both sides of the above equation. The values of 'K' and 'n' obtained will be the average values for the two different concentrations used in the calculation.

Procedure:

Make up around 100 ml of each solution of approximately 1N, 0.5 N and 0.25N acetic acid. Mix 25 ml of each acetic acid solution with 25 ml of carbon tetrachloride in three 60 ml separatory funnels and immerses into a 25°C constant temperature bath.

After 20 minutes, with regular shaking, remove the funnels and carefully draw off most (not all) of the lower (carbon tetrachloride) layers into three 100 ml Erlenmeyer flasks. Immediately sample 20 ml, using 10 ml pipette, into an Erlenmeyer flask and dilute with 100ml of water. Titrate each samples with 0.01N NaOH, using phenolphthalein as an indicator.

Similarly 10 ml sample of the water layer is pipetted from each group and titrated with 0.1N NaOH, using phenolphthalein as an indicator.

Note that either a safety bulb or a very long piece of tubing should be used with the pipettes to minimize the danger of poisoning due to the carbon tetrachloride.

Calculation:

Calculate the average K_a and n from the 1N and 0.5N acetic acid solutions and from the 0.5N and 0.25N solutions, in all cases choose water as solvent 2.

Results and Discussion:

The dissociation of weak acid increases with dilution, while the association decreases. Due to the high concentrations of acetic acid used in the system, there will be a slight dissociation of acetic acid in the water layer. Using this as a basis, decide whether or not acetic acid is associated or dissociated at these concentrations in the above solvents.

Initial Conc. of CH ₃ COOH (N)	CCl ₄ LAYER						WATER LAYER	
	Vol. of CCl ₄ (mL)	Vol. of NaOH (mL)	Conc. of NaOH (N)	Conc. of CH ₃ COOH	Vol. of WATER (mL)	Vol. of NaOH	Conc. of NaOH	Conc. of CH ₃ COOH
1.0 N	20		0.01			10		0.1N
0.5 N	20		0.01			10		0.1N
0.25 N	20		0.01			10		0.1N

Initial Conc.
of CH₃COOH

K_c

N

1.0 N – 0.5 N

0.5 N – 0.25 N

Average _____

EXPERIMENT IX MICELLAR SOLUBILIZATION

Objective:

To determine the solubilizing action of Tween 20 on Peppermint oil in water.

Background:

Micelles, the association of surfactants, increase the solubility of materials that are normally insoluble or only slightly soluble in the dispersion medium and this phenomenon is known as solubilization. Solubilization has been used in pharmacy to bring into solution a wide range of materials including volatile oils, coal tar and resinous materials, Phenobarbital, sulfonamides, vitamins, hormones and dyes.

The pharmacist must give due attention to several factors when attempting to formulate solubilized systems successfully. It is essential that, at the concentration employed, the surface active agent, if taken internally, be non toxic, miscible with solvent (usually water), compatible with the material to be solubilized, free from disagreeable odor and taste and relatively non volatile.

The amount of surfactant used is important. Large excess is undesirable both from the point of view of possible toxicity and reduced absorption and activity and an insufficient amount can lead to precipitation of the solubilized material. The amount of surfactant required to solubilize is a function of hydrophilic-lipophilic balance of the surfactant.

In this experiment, we shall see the solubilizing action of Tween 20 on peppermint oil in water. On gradual addition of water to a 50:50 mixture of peppermint oil and Tween 20 the system changed from a homogeneous mixture to viscous gel. On the further addition of water, a clear solution again formed which then separated in to two layers. As the system contains three components, the results may be best presented in the form of ternary diagram.

Procedure

Prepare 20 ml of the following three mixtures containing equal amount of peppermint oil and Tween 20 water and Tween 20 and peppermint oil and water. Titrate each mixtures with the remaining component of the system and observe the physical change. Calculate the concentration and draw the ternary diagram.

EXPERIMENT XIX

Rheology-Determination of Viscosity

Objective:

To determine the effect of temperature on viscosity; to determine the flow property of starch mucilage using rotational viscometer

Background:

Rheology is the study of the flow of fluids and the deformation of solids. Resistance is offered when one part of a fluid (liquid or gas) is moved past another. The force required to slip one layer of a fluid past another with a given viscosity is called the shearing stress while the rate of movement is called the rate of share. Resistance to this movement is called the viscosity. The viscosity of a liquid can therefore be defined as the force per unit area necessary to maintain a unit velocity between two parallel planes of the liquid separated by a unit distance. This mathematically is written as the viscosity is equal to the force per unit area (F/A) (dynes/sq cm) /velocity gradient perpendicular to the planes per second (dv/dx). The units of viscosity are poise. Since most pharmaceutical liquids have relatively low viscosities the units are often centipoises (cps) or 0.01 poise. The new units internationally are millipascals or mPc. They have the same numerical value as a cps so that one mPc is equal to one cps.

Pure liquids and true solutions have a constant viscosity at a given temperature and pressure. Such fluids are known as Newtonian fluids. The Viscosity of these fluids decrease with increased temperature. This suggests the existence of molecular clustering or association in liquids. An increase in temperature or kinetic energy disrupts these associations with a resulting decrease in viscosity. In liquid polyphasic pharmaceuticals (dispersions of gums or other polymers) the rate of share is not directly proportional to the shearing stress and the viscosity is not constant. Such fluids are called now-Newtonian fluids.

The Selection of a viscometer has a paramount importance in determining the rheology of different systems. Basically there are two classes of viscometers, one point viscometer, characterized by developing a single share stress and so a single shear rate and multiple point viscometer, which operate at varieties of rates of shear. Therefore it is possible to use both kinds of viscometers with Newtonian systems. But determining the viscosity of a non-Newtonian

system is restricted to the multiple ones, for this plane reason, the direct proportionality between shear rate (G) and share stress (F) no more holds for such systems.

1/ Determination of the effect of temperature on viscosity using Ostwald viscometer

Absolute viscosity can be measured directly by measuring the time required for a measured volume of liquid to flow through a capillary tube under an applied pressure. The experimental determination of absolute viscosity is a difficult task. The measurement of relative viscosity is an easy task. Therefore, one can measure relative viscosity by using reference fluids with known viscosities. An Ostwald viscometer will be used.

With the Ostwald viscometer the pressure driving the liquid through the capillary depends on the difference in the liquid level (h) and the density of the liquid (ρ) and the acceleration due to gravity (g). This is often called hydrostatic pressure. This pressure represents the shearing stress. The rate of flow of the liquid through the capillary represents the rate of shear. Therefore, the viscosity of a liquid is proportional to the hydrostatic pressure divided by the volume of fluid that flows through the capillary per unit time. If you use the same viscometer to make the measurements than the terms h & g and Volume are constant and you can establish that the ratio of the viscosity of one liquid to the viscosity of the second is equal to the ratio of the time it take to pass through the viscometer times the density of the liquid for the two liquids.

$$\text{Viscosity one} / \text{Viscosity two} = (\rho_1 \times \text{Time one}) / (\rho_2 \times \text{Time two})$$

To obtain good results the apparatus must be clean and dry.

Procedure:

Clamp the Ostwald viscometer in the thermostated bath so that the upper fiducial mark is below the bath surface so that the meniscus of the solution in the viscometer is visible. Pipette an appropriate amount (the exact amount will depend on the viscometer) of the solution to be studied, acetone, into the viscometer and wait until the temperature equilibrated (the amount pipetted should be the same in all trials). Draw the solution above the upper fiducial mark using mouth suction through a piece of rubber tubing and time the fall of the solution meniscus between the upper and lower fiducial marks to ± 0.01 sec. Repeat this procedure until three trials are obtained. Conduct the experiment at 70°F, 80°F, 90 °F and 100 °F. Sketch temp vs. viscosity graph for acetone and discuss the effect of temperature on its viscosity.

Temp (°F)	32	40	50	60	70	80	90	100	120	140	160	180	200
Visc (Cps)	1.794	1.546	1.310	1.129	0.982	0.862	0.764	0.682	0.559	0.470	0.401	0.347	0.305

II/ Determination of the flow properties of starch mucilage using rotational viscometer

Usually colloids of polymers like gum acacia, gum tragacanth, CMC e.t.c. are found to display a pseudoplastic flow. The relationship between the shear stress F and the shear rate G for such systems is given by

$$\eta = F^N / G \quad \text{where, } N \neq 1 \dots\dots\dots \text{eq.2}$$

Procedure:

Determine the viscosity of starch mucilage at different rpm (10, 20, 50, 100 and 200) using a rotational viscometer and then draw the log G (rpm) vs. log F (apparent viscosity) plot. Determine the flow property of the dispersion system. (Don't forget to record the temperature at which the experiment is conducted).

Results:

Temperature (°C)	Rpm	10	20	50	100	200
	Viscosity (cps)					

Discussion:

- Briefly explain the pharmaceutical importance of determining the rheological properties of liquids.
- For what kind of systems can we use one point viscometer? What is the typical property of Newtonian systems?
- According to your result, what is the effect of temperature on acetone viscosity? Explain why?
- Calculate the absolute viscosity (η) of starch mucilage at zero shear rate. What is the value of N (according to equation 2) for 6% starch mucilage at a given temperature?